

Comparative Analysis: Existing SMS Spam Detection Models vs. Our Proposed System

Mini Hackathon 2 Report
NLP Specialization Capstone Project

1. Objective

Mini Hackathon 1 established the design of our proposed **Intelligent Spam SMS Detection** system. The Review 2 report presents the implemented system, evaluates its performance, and compares the proposed architecture with traditional machine-learning approaches and recent SMS spam detection methods.

The main objective is to investigate whether combining **Semantic, Pattern, and Context Experts** through **Dynamic Evidence Routing** provides a practical approach for SMS spam classification while also providing interpretable evidence weights for individual messages.

2. Evolution of Approaches in the Literature

SMS spam detection has evolved from statistical machine-learning methods toward contextual transformer-based and specialized deep-learning approaches.

Generation	Representative Approach	Main Focus
1. Traditional ML	TF-IDF + Logistic Regression, TF-IDF + SVM	Word-frequency and sparse textual features
2. Sequence Deep Learning	CNN, RNN, LSTM, BiLSTM	Sequential text patterns
3. Transformer-based Detection	BERT, RoBERTa, DistilBERT	Contextual language understanding
4. Specialized Hybrid Models	BERT + GCN, Fuzzy + BiLSTM, etc.	Combining multiple representations
5. Explainable / Robust Detection	XAI, adversarial and augmentation-based methods	Interpretability and robustness
6. Proposed System	DistilBERT + Semantic/Pattern/Context Experts + Dynamic Router	Instance-specific evidence weighting and fusion

3. Comparative Analysis: Existing Models vs. Proposed System

Model / Work	Core Approach	Main Purpose	Key Features	Dynamic Routing	Dataset / Domain	Relation to Our Project
TF-IDF + Logistic Regression	TF-IDF + Logistic Regression	SMS classification (Spam vs Ham)	Bag-of-words, word-frequency features	✗	SMS Spam Collection (UCI)	Traditional baseline using statistical text features.
TF-IDF + Linear SVM	TF-IDF + Support Vector Machine	SMS classification (Spam vs Ham)	Sparse textual representation, high-dimensional features	✗	SMS Spam Collection (UCI)	Strong traditional baseline for comparison.
CNN for Text	Convolutional Neural Network (CNN)	Spam detection using n-gram features	Captures local patterns and key phrases	✗	SMS Spam Collection (UCI)	Uses deep learning for pattern extraction.
LSTM / BiLSTM	Recurrent Neural Network (LSTM / BiLSTM)	Spam classification using sequential text	Captures long-term dependencies in text	✗	SMS Spam Collection (UCI)	Models sequential information in SMS messages.
BERT-based Models	BERT / RoBERTa (fine-tuned)	Contextual spam detection	Contextual language representations	✗	SMS Spam Collection (UCI)	Uses transformer models but single representation (no expert routing).
BERT-G3CN (2025)	BERT + Graph Convolutional Networks	Advanced spam detection	Combines textual and graph features	✗	SMS Spam Collection (UCI)	Uses graph-based evidence rather than dynamic expert routing.
Fuzzy + BiLSTM (2025)	Fuzzy Logic + BiLSTM + Attention	Spam classification	Combines fuzzy linguistic features with sequence modelling	✗	SMS Spam Collection (UCI)	Uses fuzzy rules and attention but does not use dynamic routing.
Our Proposed Model (2026)	DistilBERT + Semantic / Pattern / Context Experts + Dynamic Router	Spam classification (Spam vs Ham)	Multiple evidence experts, instance-specific weighting and fusion	✓	SMS Spam Collection (UCI)	Dynamically combines semantic, pattern and contextual evidence for each SMS message.

4. Performance Comparison Across Studies

The proposed model was evaluated on a **517-message unseen test split** from the SMS Spam Collection dataset. Traditional machine-learning baselines were evaluated on the same test split, making these comparisons directly comparable within our experimental setup.

Model	Accuracy	Precision	Recall	F1-Score
TF-IDF + Logistic Regression	96.32%	100.00%	70.77%	82.88%
TF-IDF + Linear SVM	98.07%	96.61%	87.69%	91.94%
Our Proposed Model	99.42%	100.00%	95.38%	97.64%

The proposed model achieved **99.42% accuracy, 100% precision, 95.38% recall, and 97.64% F1-score** on the 517-message test split.

Published results from other SMS spam detection studies use different datasets, splits, and evaluation protocols. Therefore, their reported accuracy values are presented separately as literature context rather than as a directly comparable leaderboard..

6. Key Takeaways

- **Strong classification performance:** The proposed model achieved **99.42% accuracy and 97.64% F1-score** on the 517-message test split.
- **Multiple evidence representations:** The architecture combines **Semantic, Pattern, and Context Experts** rather than relying on a single textual representation.
- **Dynamic routing:** The Dynamic Evidence Router assigns **instance-specific weights** to the different experts for each SMS message.
- **Baseline evaluation:** TF-IDF + Logistic Regression and TF-IDF + Linear SVM were evaluated using the same test split, providing directly comparable traditional baselines.
- **Practical deployment:** The trained model was integrated into a **Flask web application**, allowing users to enter SMS messages and obtain SPAM/HAM predictions.
- **Evidence information:** The web application can display the model confidence and the relative weights assigned to the Semantic, Pattern, and Context Experts.
- **Controlled robustness test:** The model maintained its predictions for all **5 controlled perturbation samples**, giving a **100% prediction-preservation rate on those five samples**. This is a small controlled evaluation and should not be interpreted as proof of general robustness.
- **Literature comparison:** Recent approaches use different techniques such as graph convolution, RoBERTa-based explainability, and fuzzy sequence modelling. The proposed system explores a different architectural combination based on dynamic evidence routing.

7. Research Gap Addressed

Existing SMS spam detection approaches use a variety of techniques, including statistical text features, recurrent networks, transformers, graph-based representations, fuzzy logic, data augmentation, and explainability methods.

The proposed system addresses the research gap by exploring a **multi-expert evidence architecture** in which different aspects of an SMS are processed separately and then dynamically combined:

SMS → DistilBERT → Semantic / Pattern / Context Experts → Dynamic Evidence Router → Dynamic Evidence Fusion → SPAM/HAM

Instead of assigning a fixed contribution to each representation, the router produces **instance-specific evidence weights** for each SMS.

The project therefore focuses on the combination of **contextual language understanding, specialized evidence extraction, and dynamic evidence fusion** within a practical SMS spam detection system.

8. Conclusion

The Mini Hackathon 2 implementation demonstrates a complete **Intelligent Spam SMS Detection** pipeline using **DistilBERT, Semantic/Pattern/Context Experts, Dynamic Evidence Routing, and Dynamic Evidence Fusion**.

On the 517-message test split, the proposed model achieved **99.42% accuracy, 100% precision, 95.38% recall, and 97.64% F1-score**. The model was also integrated into a Flask-based web application capable of producing **SPAM/HAM predictions, confidence values, and dynamic evidence weights**.

The primary architectural contribution of the project is the use of **instance-specific Dynamic Evidence Routing** to combine complementary semantic, pattern, and contextual information for SMS spam classification. The system provides a practical research prototype that can be extended with larger datasets, additional experts, alternative routing mechanisms, and further robustness evaluation.

Prepared for Mini Hackathon 2, building on the literature survey and the Mini Hackathon 1 design report.